AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM240
Introduction to Computer Graphics

Midterm Examination (MTA)
Spring semester 2024/2025

Date:9 /April/ 2025

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Number of Exam Pages: 3 (including this cover sheet)	Time Allowed: 90

Instructions:

- Total Marks: **60**
- This exam consists of **2** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Part -I: Question 1: Choose the best correct answer of all the following: (15 Marks)

1 ..... is used to define visible colors with numerical values

A Alpha channel
B Color space
C Rendering
D None of the above

2 The function draw which is a function in the graphics2D object is an example of .....functions

A Transformation
B Rendering
C Modeling
D None of the above

3 The function LineTo has .....parameters.

A 4
B 3
C 1
D 2

4 An alpha-value is a number between 0.0 and 1.0 with .....representing complete transparency .

A 1
B 0.5
C 10
D 0

5 The center of the rectangle defined by RoundRectangle2D.Double(a1,a2,a3,a4,a5,a6) is at .....

A (a1+0.5a3,a2+0.5a4)
B (a1+0.5a5,a2+0.5a6)
C (a5,a6)
D (a3,a4)

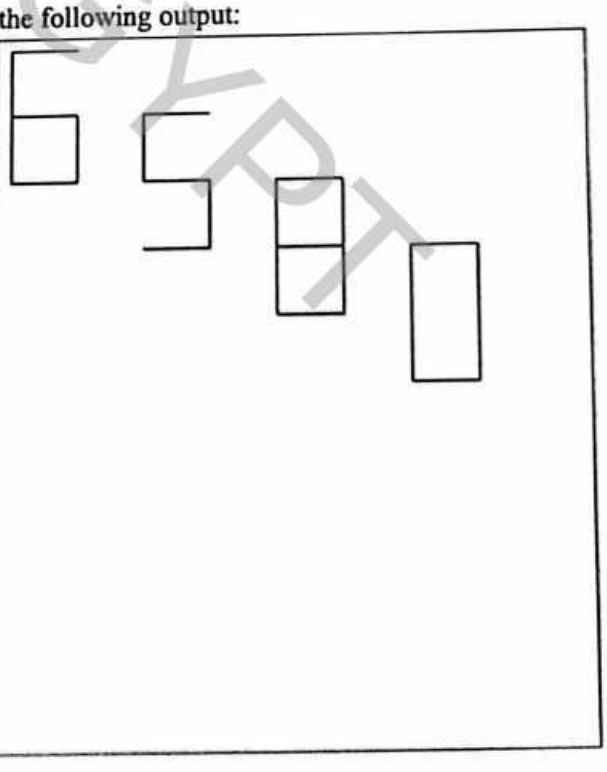
Part-II: Answer all of the following questions (45 Marks).

Question 2: (15 marks)

a. Explain the concept of approximation in the context of 3D modeling and discuss the benefits it offers.

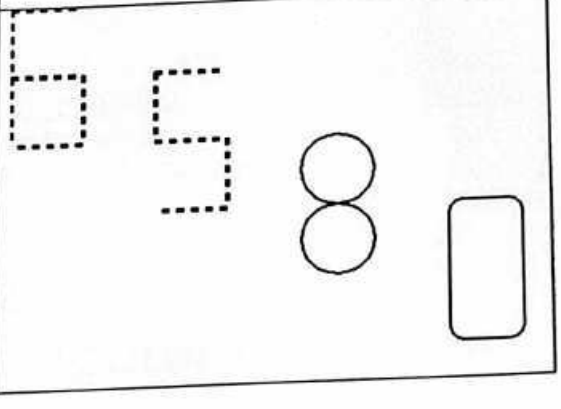
b. Having the following Java code along with the following output:

```
g.drawLine(0,0,50,0);
g.drawLine(0,0,0,100);
g.drawLine(0,50,50,50);
g.drawLine(50,50,50,100);
g.drawLine(50,100,0,100);
g.drawLine(100,50,150,50);
g.drawLine(100,50,100,100);
g.drawLine(100,100,150,100);
g.drawLine(150,100,150,150);
g.drawLine(150,150,100,150);
g.drawLine(200,100,250,100);
g.drawLine(200,150,250,150);
g.drawLine(200,200,250,200);
g.drawLine(200,100,200,200);
g.drawLine(250,100,250,200);
g.drawLine(300, 150, 350, 150);
g.drawLine(350, 150, 350, 250);
```



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g.drawLine(350, 250, 300, 250); g.drawLine(300, 250, 300, 150);	
Update the above code to output the following shapes:	

Question 3: (15 marks)

a. Explain the concept of control points in curve definition and how they influence the shape of the curve.

b. Having the following shape:



Write a java code segment to plot it.

Question 4: (15 marks)

a. Explain the concept of a clipping path and its role in image rendering.

b.

Having the following code in Java: Shape shp=new RoundRectangle2D.Double(100,100,59,26,5,5); The code has the following output:	

Draft the output when adding the following code to the above code:

AffineTransform af=new AffineTransform();

af.rotate(4.574428, 159, 126);

End of questions

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